Reg. No. | | | | | | | | | | |

[14ME63]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

COMPOSITE MATERIALS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions, choosing at least **TWO** questions from each part
2. Missing data, if any, may be suitably assumed.

PART - A

1.	a.	Classify and explain composite materials on the basis of matrix and reinforcement	08
	b.	Highlight the advantages and applications of fibre reinforced and laminated composites.	06
2.	a.	Derive an expression for elastic modulus of composites.	10
	b.	Write a note on Rule of Mixture in composites.	04
3.	a.	With the help of neat sketch explain pultrusion process and filament winding.	10
	b.	Differentiate between manual cutting and computer aided cutting.	04
4.	a.	With the help of neat sketch explain following composite manufacturing process;	14
		i. Thermoforming process	
		ii. Injection moulding process	
		iii. Bag moulding	

PART - B

5.	a.	Discuss the various factors to be considered for selection of matrix and reinforcement materials in MMCs.	08
	b.	Highlight the advantages, limitations and applications of MMCs.	06
6.	a.	Differentiate between powder metallurgy and liquid metallurgy technique used for fabrication of MMCs.	08
	b.	Explain the importance of secondary process in fabrication of MMCs.	06
7.	a.	Explain thermoset and thermoplastic polymers with examples.	08
	b.	Discuss properties and applications of polymer matrix composites.	06
8.	a.	With the help of neat sketch explain any one technique used for synthesis of nano composite.	06
	b.	What are nano composites? Explain its characteristics and applications.	08

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